

Student Answer Sheet

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Subject: Science | Class: 10A | Practice Test

Q1:

Photosynthesis is process where green plants convert light energy into chemical energy using CO₂ and H₂O producing glucose (C₆H₁₂O₆) and O₂. Happens in chloroplasts using chlorophyll.

Q2:

Newton Second Law: The rate of change of momentum of a body is directly proportional to applied force. $F = ma$ where F=force, m=mass, a=acceleration.

Q3:

DNA is deoxyribonucleic acid with double helix structure. Two strands held by hydrogen bonds between complementary bases: Adenine-Thymine and Guanine-Cytosine. Found in nucleus carrying genetic information.

Q4:

Mitosis: 2 identical daughter cells, diploid (2n), somatic cells, growth repair. Meiosis: 4 genetically diverse cells, haploid (n), germ cells, sexual reproduction. Meiosis has two divisions unlike mitosis.

Q5:

Water cycle stages: Evaporation (solar energy converts liquid to vapor), Transpiration (plants release vapor), Condensation (vapor forms clouds), Precipitation (rain/snow), Collection (rivers, groundwater, oceans). Continuous process.

Q6:

Human heart: 4 chambers (RA, RV, LA, LV). Right side: deoxygenated blood to lungs. Left side: oxygenated blood to body. SA node controls heartbeat. Average 72 bpm. 4 valves prevent backflow.